

Subscripts and Superscripts

Example 1 : Basic Subscripts

Subscripts are written with underscore:

$$x_1 \in S$$

$$x_2 = x_1 + 1$$

Common for indexed variables in sequences, arrays, or iterations.

Example 2 : Basic Superscripts (Exponentiation)

Superscripts denote exponentiation:

$$x^2 = x * x$$

$$n^3 = n * n * n$$

The caret symbol creates superscripts for powers.

Example 3 : Multiple Subscripted Variables

$x_1 : \mathbb{N}$
$x_2 : \mathbb{N}$
$x_3 : \mathbb{N}$
$x_1 = 1$
$x_2 = 2$
$x_3 = 3$

Define multiple indexed variables.

Example 4 : Sequence Indexing

Subscripts are natural for sequence element notation:

$$\forall i : 1 \dots n \bullet s_i \in \mathbb{N}$$

Here s_i represents the i-th element of a conceptual sequence.

Example 5 : Exponentiation in Expressions

$$\{ n : \mathbb{N} \mid n < 10 \bullet n * n \}$$

Set of squares. Z/fuzz has no numeric exponentiation operator, so a set comprehension body writes the square as n * n rather than n^2.

Example 6 : Power Function for Nested Exponents

For nested exponents like power of a power, define a power function:

$pow : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$
$\forall x : \mathbb{N} \bullet pow(x, 0) = 1$
$\forall x, n : \mathbb{N} \bullet n > 0 \Rightarrow pow(x, n) = x * pow(x, n - 1)$

Then use it for nested exponentiation:

$$pow(pow(x, 2), 3) = pow(x, 6)$$

$$pow(pow(n, 2), n + 1)$$

This avoids LaTeX double superscript issues.

Example 7 : Polynomials

Subscripts for coefficients, superscripts for powers:

$$a_0 + a_1 * x + a_2 * x^2 + a_3 * x^3$$

Standard polynomial notation.

Example 8 : Combined Subscripts and Superscripts

Variable with subscript, raised to a power:

$$x_{-i}^2$$

Meaning: x sub i, squared.

Example 9 : Cartesian Product Powers

Cartesian product repeated n times:

$$S^2 = S \times S$$

$$S^3 = S \times S \times S$$

Set of n-tuples from S.

Example 10 : Iteration Subscripts

Subscripts often denote iteration steps:

$$x_0 = initial$$

$$x_{-}(i + 1) = f(x_{-}i)$$

Defines a recurrence relation.

Example 11 : State Variable Subscripts

<i>Counter</i>	
<i>count</i> : $\mathbb{N}$	
<i>countNext</i> : $\mathbb{N}$	
<i>countNext</i> = <i>count</i> + 1	

Next-state variables can use descriptive names like countNext instead of primed notation.

Example 12 : Indexed Family of Sets

$S_1 : \mathbb{P} \mathbb{N}$
$S_2 : \mathbb{P} \mathbb{N}$
$S_3 : \mathbb{P} \mathbb{N}$
$S_1 = \{1, 2, 3\}$
$S_2 = \{4, 5, 6\}$
$S_3 = \{7, 8, 9\}$

Family of sets indexed by natural numbers.

Example 13 : Exponents in Constraints

$$\forall n : \mathbb{N} \bullet n^2 \geq n$$

Constraint using exponentiation: every natural number's square is at least itself.

Example 14 : Complex Subscript Expressions

$$x_{-}(i+1) + x_{-}(i-1) = 2 * x_{-}i$$

Recurrence relation with subscript expressions.

Example 15 : Power Set Notation

Sometimes written with superscript:

$$\mathbb{P} S \vee 2^S$$

Both denote the power set of S.

Example 16 : Best Practices

Guidelines for subscripts and superscripts: 1. Use subscripts for indices, versions, or families 2. Use superscripts for exponents and powers 3. For nested exponents, define a pow function 4. Be consistent in indexing: start at 0 or 1, not mixed 5. Document what subscripts mean